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Published by Oxford University Press, Inc.
198 Madison Avenue, New York, New York 10016

www.oup.com

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Library of Congress Cataloging-in-Publication Data
The science and psychology of music performance : creative strategies for teaching and
learning / Richard Parncutt and Gary E. McPherson, eds.

p. cm.

Includes bibliographical references and index.

ISBN 0-19-513810-4

1. Music—Performance—Psychological aspects. 2. Music—Instruction and study.

I. Parncutt, Richard, 1957– II. McPherson, Gary E.

ML3838 .S385 2002

781.4—dc21 2001036292

1 3 5 7 9 8 6 4 2

Printed in the United States of America
on acid-free paper

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Wind Instruments

LEONARDO FUKS & HEINZ FADLE

In mouth-blown wind instruments, the energy provided by the respiratory system is converted directly into sound. In all cases a primary vibrating element, generically called *reed*, controls the airstream. The reed may be a piece of bamboo, the lips, a metallic tongue, or even the air jet (in flutes and recorders). Players control loudness, attack, intonation, and timbre by means of embouchure settings, blowing pressure, airflow, and length of the air column. The respiratory muscles perform complex and systematic movements, generating wide ranges of pressures, and coordinated oscillations that produce the vibrato effect. Intonation may be affected by the characteristics of the lung air. We address the associated sensory, physiological, and acoustical phenomena. Common controversial or misleading concepts among wind players are discussed and some simple experiments are proposed for pedagogical applications.

Mouth-blown wind instruments convert pneumatic energy, in the form of air pressure and velocity, into sound waves. They belong to the category that Sachs (1940) describes as *aerophones*. Other aerophones include the human voice, the pipe organ, and the accordion.

Wind players use their respiratory apparatus as an air compressor to supply energy to the instrument. At the mouth, air is delivered through a sophisticated interface with the instrument, which musicians refer to as their *embouchure*. Each subtle action from muscles of expiration and embouchure imprints the sound, forming part of the musical expression. The arms, hands, and fingers skillfully control a complex of tone holes, slides, rotating valves, levers, and other mechanisms. Inside the instrument, several phenomena take place, including reflection, diffraction, resonance, shock waves, damping, vortex generation, and chaotic oscillations. Body posture, respiratory techniques, and a psychological strategy to deal with stressful situations are additional key aspects of perfor-

mance. This rich environment challenges the abilities of players and educators to deal with a number of principles that are associated with performance on wind instruments. Some of the most important are described in this chapter.

The Reed

Every wind instrument has some mechanism that cyclically disturbs or interrupts the air stream. This mechanism can be generically called a *reed*. Therefore, rather than a component of bamboo or other elastic material, *reed* in this context stands for a valve that controls the air passing through the instrument, as shown in Figure 20.1. This concept has been consistently adopted by several authors in music acoustics (Campbell & Greated, 1987; Sundberg, 1991; Strong & Plitnik, 1992). The tongue, the lips, and the moving structures of the respiratory system may also be used to regulate the airflow, serving as additional valves.

Winds may incorporate a *mechanical reed* (reed woodwinds), *free reed* (mouth harmonicas), *lip reed* (brass instruments), or *air reed* (flutes and recorders), where one side is in contact with the player's mouth and the other attached to the instrument. Figure 20.1 shows four simplified models of reeds, each one representing a different type of oscillation. Because instruments such as the flute involve an air reed with no solid vibrating part, they are discussed separately.

As the air stream encounters resistance in passing the reed, a difference of pressure will be created between both sides. The pressure in the upstream (before the reed) is in principle higher than that of the downstream side due to friction losses. As the reed is flexible, it will tend to bend or move due to this pressure difference. In much the same way as a spring, a reed's elasticity will allow it to return to its original position and shape. The reed also contains mass that is distributed along its body. This defines a *spring-mass* system that has its own resonance frequencies.

In a similar manner to a tuning fork, a reed can be excited to oscillate at its own frequencies. This is the case in free reeds and, exceptionally, in clarinets

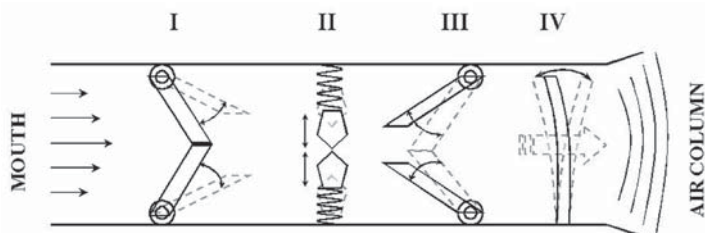


Figure 20.1. Oscillation types found in lip, mechanical, and free reeds: rest position of the reed (solid line) and extreme position during vibration (broken line). (I) outward-striking and (II) sideways- or upward-striking oscillations as found in brass instruments; (III) inward-striking oscillation, typical of reed woodwinds; and (IV) free-striking oscillations, which occur in free reeds.

that produce the so-called *squeak sounds* (Bouasse, 1929/1986). However, the tissue of the lips, forming a cushion pressed against the reeds, absorbs most of the reed energy in every cycle, with the oscillations being damped more than those of a tuning fork.

Another factor is the so-called *Bernoulli force*. When air passes through a narrow passage, the particles' velocity must increase in order to maintain a constant flow. This results in a drop in pressure past the narrow passage and a subsequent partial vacuum between air molecules. This negative pressure tends to suck the reed toward the closing position. If the closing reed blocks the air, this suction is immediately interrupted, releasing the reed to return to its original state.

In spite of being easily explained in general terms, the Bernoulli force is not important in all cases. It is certainly relevant in double reeds, especially the oboe, and for the higher range of brass instruments (Fletcher & Rossing, 1998). However, in the case of the clarinet its role is in most cases negligible, because it is not strong enough to make this instrument's reed vibrate (Worman, 1971).

The Air Column

The *air column* is the length of vibrating air in the instrument during sound production. The airflow that crosses the reed creates pressure waves that travel along the air column toward its end and are then reflected back to the reed. In lip and mechanical reeds, if the reflected wave pressure is strong enough it will make the reed reverse its direction. At this point, the system is ready for a new cycle, as it establishes a chain of *self-sustained oscillations*. In an air reed, the air jet flows alternately into the column and out into the ambient, as a result of the waves set up in the air column (Benade & Gans, 1968).

The connection between the air column and the reed, technically referred to as acoustical coupling, forces the reed to vibrate at certain frequencies. When producing a steady tone, the reed vibrates with a frequency close to one particular mode of the air column, so that reflected waves arrive precisely when the reed, depending on the oscillation type, closes or opens. Benade (1990) described this situation as the reed being "enslaved" by the air column. Each configuration of tone holes, keys, pistons, valves, or slide in a wind instrument determines the length and shape of the air column.

A given air column resonates at a set of frequencies that are possible inside it. Each of these frequencies corresponds to a resonance mode. The set of tones obtained with a fixed air column should ideally coincide with those of a harmonic series. However, instruments usually deviate from the perfect geometry of a cylinder or a cone, so that the real modes also deviate from the exact harmonics. Wind players must develop the skill of matching the embouchure and blowing controls to select the desired mode of the air column, which will result in particular tones. Whenever the reed and the air column achieve a pattern of self-sustained oscillations, a *playing mode* is defined. *Overblowing* happens when the instrument jumps from a mode to a higher one.

The four models in Figure 20.1 are *pressure-controlled valves*, as pressure is the main controller of the reed movement. Only air reeds such as the flute are directly controlled by airflow (Benade & Gans, 1968).

While the instrument is being played, the sound waves generated in the reed propagate both to the tube and to the mouth cavity, toward the player's airways. However, most of this energy propagated into the mouth is lost by absorption through the multiple branches and soft walls of the respiratory apparatus. Nevertheless, the vocal tract may respond to some frequencies, as it does for the formation of the vowel sounds. In such cases, the vocal tract can be considered as an internal air column that interacts with the reed and the instrument's air column.

Some skilled players attempt to master the control of the vocal tract to improve their sound quality and stability and to help in the transition between playing modes. While some authors consider that a player's vocal tract has a negligible influence on an instrument's tone (e.g., Backus, 1985, in reed woodwinds), others (Benade, 1986; Johnston, 1987; Fletcher & Rossing, 1998) claim a considerable influence of the vocal tract in instruments such as the clarinet, saxophone, harmonica, and brass.

Types of Reed Oscillation

Lip, mechanical, and free reeds perform characteristic types of oscillations according to their mechanical configuration, as shown in Figure 20.1. The airflow from the lungs crosses the reed, which is subject to forces provided by the embouchure. Helmholtz (1863/1954) referred to type I as an outward-striking and to type III as an inward-striking oscillation. Similarly, type II can be called upward- or sideways-striking and type IV a free-striking oscillation (Fletcher & Rossing, 1998).

In general, the playing frequencies in types II and III are lower than the resonant frequencies of the air column and of the reed itself, while in type I oscillations the playing frequency is higher than those of the air column and the reed alone (Fletcher & Rossing, 1998). Free reeds, corresponding to type IV, are not strongly coupled to an air column and essentially vibrate at their resonance frequencies.

The Embouchure

The *embouchure* can be defined as the constellation of forces and positions in the lips, mouth region, and face that act on the wind instrument (Porter, 1967). It consists of the muscles that form and surround the mouth, the teeth, the upper jaw and mandible, and muscles responsible for mastication. The embouchure controls the reed behavior, thus affecting tone color, articulation, dynamics, and other parameters. In defining the embouchure, it is possible to include the internal mouth configuration, such as the positioning of jaw and tongue. In this sense, the mouth cavity can be regarded as a secondary air column.

In reed woodwinds, the embouchure regulates the reed parameters, such as the stiffness, the aperture for the airflow, the degree of absorption, and, to some

extent, its effective mass (Nederveen, 1969). The embouchure also provides an airtight attachment of the lips to the instrument. In brass instruments, the lips obviously serve both as a reed and as an embouchure component and include the same functions cited earlier. In the flute, the embouchure must be skillfully shaped and managed to ensure proper generation and control of the sound. In spite of the importance of the embouchure in performance and sound production, these latter aspects seem to have been unexplored and therefore deserve more research interest.

Wind Instrument Families

Brass

Stroboscopic observations of the lip vibrations in a professional trombone player indicate that the upper lip vibrates with greater amplitudes than the lower lip in all frequencies and dynamic levels (Copley & Strong, 1996). This study also showed that on average, two-thirds of the mouthpiece diameter is covered by the upper lip and one-third by the lower lip. However, considerable variations of this proportion can be observed in the general playing population, which suggests that players tend to optimize their embouchure during the process of learning.

Lip reeds correspond to types I or II in Figure 20.1. In both cases air pressure tends to open the lips, but in type II the Bernoulli forces act alone to close the lips back. Usually, the type I occurs in the lowest playing modes, with type II being more frequent in the higher modes (Yoshikawa, 1995). It is also possible that both types of motion take place in combination, as observed across the whole compass in a trombone study (Copley & Strong, 1996). Not all studies, however, found the same behavior, probably due to differences in the measurement methods that were employed and particular techniques that were adopted by individual players (Chen & Weinreich, 1996).

A common warm-up practice among brass players consists of buzzing on the mouthpiece or with lips only. This exercise is also claimed to develop flexibility and control of lip vibrations. However, according to Yoshikawa (1995), the isolated mouthpiece represents a completely different acoustical system from the full instrument bore, supporting only type I oscillations but not type II. This finding suggests that exercises of this type do not develop lip flexibility in the way often assumed by brass teachers.

The vibration frequency of the lips increases with a combination of higher muscular tension and lower mass. This *mass* refers to the section of the lip that effectively vibrates. Also, raising the pitch requires higher blowing pressure (Bouhuys, 1964). Therefore, it could be expected that the player would tend to press the lips more firmly against the mouthpiece when playing higher tones. This was corroborated by Barbenal, Davies, and Kenny (1986), who observed that the compressive forces between lips and mouthpiece increase systematically with increasing pitch in professional players. Once again, this finding contradicts some

performers' claim that it is possible to develop a rather "loose" and "fixed" embouchure, regardless of pitch and dynamic level. However, the idea of a loose embouchure may work as a mental image that helps to achieve the goal of minimizing the effort and strain of the embouchure muscles in order to preserve the necessary sensory response of the tissues.

Reed Woodwinds

Mechanical reeds correspond to type III oscillations in Figure 20.1, where the airflow tends to blow them closed. In the oboe and in the bassoon, the reeds beat against each other almost all the time, completely interrupting the airflow in every oscillation. Similarly, in the saxophone, the reed tends to repeatedly beat against the mouthpiece. This occurs because the conical shape of those instruments creates comparatively high pressures where the reeds are located. In the clarinet, however, with its cylindrical shape, the reed does not necessarily beat against the mouthpiece for the relatively soft tones (Worman, 1971). This explains why softer tones of the clarinet have comparatively much stronger fundamentals than louder tones.

One major distinctive characteristic among mechanical reeds is their *hardness*, also referred to as heaviness or strength. Many players agree that harder reeds need higher blowing pressure and greater lip tension. Also, harder reeds are supposed to produce darker tones with less prominent high frequencies. In a study with oboe, bassoon, clarinet, and alto saxophone (Fuks, 1998a), two contrasting reeds were used in each, one rated as hard (heavy) and another as soft (light). In all reeds and instruments, louder tones required more airflow and higher blowing pressure. Among hard and soft reeds, the harder reeds always produced louder tones, needed more airflow, and involved higher blowing pressure.

Harmonicas and Similar Instruments

Free reeds correspond to type IV oscillations in Figure 20.1. The air tends to close the free reed until it reaches the vertical line and then starts to open it. The harmonica is the most typical instrument in this category. As mentioned earlier, its usual blown frequency is about the same as if it were plucked and left to vibrate alone.

As mentioned previously, the vocal tract acts as an additional air column. The effect of the vocal tract on the sound of a diatonic harmonica can be demonstrated just by moving the tongue back and forth inside the mouth while blowing, which produces clear changes in the relative intensity of the overtones.

Air Reeds: Flutes and Recorders

Air reeds contain a jet that is directed toward an air column, which generates airflow fluctuations inside the tube. In flutelike instruments, the airflow is guided through the player's lips toward a mouth hole. In recorderlike instruments, the flow is conducted through a *flue* channel that is directed toward a sharp edge called the labium.

Two different kinds of tone can be produced in such instruments: a *pipe tone* and an *edge tone*. A pipe tone results when the jet and the air column are acoustically coupled. All pipe tones coincide with the tube's modes, which roughly belong to a harmonic series and can be shifted by overblowing. However, when the jet is relatively distant from the resonant tube acoustical coupling to the pipe is poor and the jet interacts mainly with the labium. This produces edge tones with high frequencies that are proportional to the jet velocity. Therefore, they are not discrete as pipe tones but continuously varying. Edge tones may be important during the attack of a new tone. For example, if a flute player is starting a low C_4 (247 Hz), it takes a moment before the tone is actually initiated. During this period, the edge tones sound alone, contributing to the onset of the tone and to its sound quality. Edge tones are also generated when the pipe is damped (filled with cotton) and when the pipe is simply removed (Verge, 1995).

Air reeds are *flow-controlled* systems in which the pressure at both ends of the air column is close to the ambient pressure. For that reason, the air particles reach higher velocity close to these ends, so that they are directly affected by the airflow. Consequently, the player's vocal tract is usually weakly coupled to the flute and therefore has only a limited influence on the sound (Coltman, 1973; Fletcher & Rossing, 1998).

Boehm (1871/1964), the inventor of the modern flute, explains that the air stream strikes the sharp edge of the mouth hole and is broken (or divided) so that one part goes over or beyond the hole while the other part produces the tone. This is not a perfect description of what really happens, but it is still to be found in many textbooks. A flute can even produce a pipe tone when the edge of the mouth hole is dull. This can be demonstrated by applying a thin layer of wax or chewing gum to the edge of the mouth hole. If the wax or gum is placed in the right position, sound production might still be possible. In this case, however, the edge tones may be hampered, thus affecting the attack and overall sound quality.

Techniques

Using the Respiratory Muscles

The diaphragm is a large and thin muscle group located at the base of the lungs. It separates the thorax from the abdomen. Contrary to common belief, the diaphragm is solely an inspiratory agent that contracts when one breathes in by descending its central part, drawing the lungs downward, increasing the longitudinal dimension of the chest, and elevating the lower ribs (Roussos, 1995). These actions produce negative pressure around the lungs. During wind playing, blowing pressures range from almost 0 (e.g., *piano* low tones in the flute) to values that exceed 120 cm H_2O in *fortissimo* high-pitched trumpet tones (Bouhuys, 1968; Pawłowski & Zółtowski, 1987; Fletcher & Tornapolsky, 1999). If a player completely fills the lungs and closes the airways, without any expiratory effort, a pressure between 30 and 50 cm H_2O will be achieved. This is called relaxation pressure and is generated by the elasticity of the respiratory system.

Similarly, if the lungs are only partially filled—to 70% of their capacity, for example—the relaxation pressure will not exceed 20 cm H₂O (Agostoni & Mead, 1964).

Whenever pressures higher than the relaxation pressure are needed, the expiratory muscles (abdominal and internal intercostal muscles) will be used. However, if pressures lower than the relaxation values are necessary, the inspiratory muscles (external intercostal muscles and the diaphragm) will be used (Bouhuys, 1968). An example of the latter is playing the recorder at high lung volume, when the relaxation pressure is relatively higher than the required blowing pressure. This may also happen at some ranges of the flute and other instruments. Diaphragm activity in skilled flutists has been assessed in at least three studies (Gärtner, 1973; Cossette, 1993; Cossette, Sliwinski, & Macklem, 2000), and all of them indicate that some players make systematic use of the diaphragm, while others do not use it most of the time. These findings suggest that excessive emphasis has been given to the diaphragm in the description of wind performance.

Blowing Pressures

The pressure measured inside the player's mouth during performance is a major input parameter that has been documented in a number of studies. In reed woodwinds, blowing pressure is intimately related to airflow. Acoustical and aerodynamical considerations show that in reed woodwinds the oscillations initiate at a minimal pressure of approximately one-third of the pressure that completely closes the reed. In clarinets, for a fixed embouchure, the airflow and the sound power radiated by the instrument tend to decrease with increasing blowing pressure (Worman, 1971). If the player keeps a constant embouchure while increasing blowing pressure, the tone gets softer until eventually the reed simply closes. This apparent discrepancy explains why players must reduce the embouchure tightness when playing a *crescendo*.

Fuks and Sundberg (1999a) found that blowing pressures in the oboe, bassoon, clarinet, and alto saxophone are systematic in professional players. These authors measured the variation of blowing pressure with loudness and fundamental frequency at four dynamic levels: in isolated tones and musical arpeggios performed by two players of each instrument. Figure 20.2 displays part of the results, where each instrument presents characteristic curves, differing only slightly between players.

Vivona (1968) observed that nonprofessional trombone players use blowing pressures that increase with pitch and dynamic level. For tones B₂, B₃, and B₄, the average pressures were 11, 16, and 39 cm H₂O in “soft volume level” and 24, 39, and 67 cm H₂O for “loud levels.” This behavior is similar to reed woodwinds and should also apply to other brass instruments at different ranges.

A study of experienced trumpet players (Fletcher & Tornapolsky, 1999) showed that the minimal blowing pressure (i.e., threshold pressure) increased proportionally with the frequency of the tone played. In the first range, starting from threshold pressure, each doubling of blowing pressure caused the sound pres-

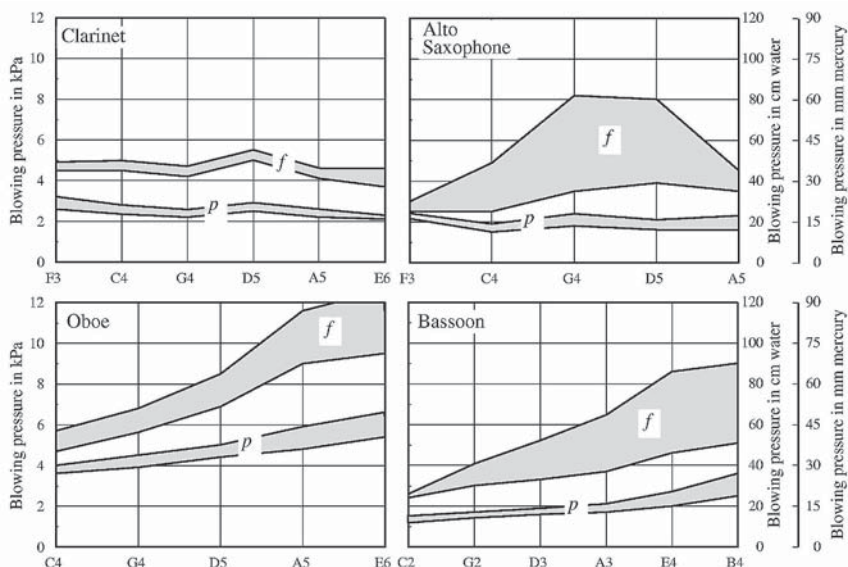


Figure 20.2. Blowing pressures in clarinet, alto saxophone, oboe, and bassoon tones, played at different dynamic levels by two professionals on each instrument. Pressure values are marked in kilopascal on the vertical axis on the left and in cm H₂O and mm of mercury on the right. In all instruments, blowing pressures increased from *piano* to *forte*. In the oboe and bassoon, the pressure systematically increased with fundamental frequency. In the clarinet, there was a slight decrease of pressure with fundamental frequency. In the saxophone, the pressure tended to increase with pitch in the first octave and followed the clarinet in the second octave. (From Fletcher & Rossing, 1998, after data from Fuks & Sundberg, 1999a. Copyright Springer-Verlag 1998, used by permission.)

sure level to increase by about fifteen decibels, whereas in a second and higher range, called the saturation region, doubling of blowing pressure increased the sound level by only three decibels. In some players, extreme pressures as high as 25 kPa (250 cm H₂O) were measured.

In reed woodwinds, there is also an important relationship between blowing pressures and intonation. As a rule, as the blowing pressure is increased, the fundamental frequency rises. This can be explained by an increase in the reed's natural frequency. Some clarinet players believe that playing tones louder makes the pitch go flat, because a player automatically reduces the embouchure tension to increase loudness (as discussed earlier) and thus decreases the reed's natural frequency (Bak & Doemler, 1987). In other words, the effect of lip relaxation on fundamental frequency is greater than the effect of pressure.

A rather linear relationship between blowing pressures and fundamental frequency has been found for flutes and recorders (Herman, 1959; Coltman, 1968; Fletcher & Rossing, 1998). This means that an octave leap by overblowing will require approximately twice the pressure.

In flutes, blowing pressure varies between 2 cm H₂O for C₄ and 20 cm H₂O for C₇, with little increase between soft and loud tones (Cossette et al., 2000). A crescendo is obtained by airflow velocity increase via the reduction of the oval lip opening, in both height and width. Interestingly, professional flutists tend to use the maximal blowing pressure that a given mode can take without overblowing. They do this to ensure steadiness and quality of tone. To control the dynamic level, a player needs to change the shape and dimension of the lip opening and also move the lips forward and backward. These maneuvers affect the embouchure, the jet length, and the proportion of mouth-hole covering (Fletcher & Rossing, 1998; Cossette et al., 2000).

Perception of Pressure

The results cited earlier suggest that the sensory functions in the abdominal, thoracic, and lung regions are important to effective wind instrument playing. Also, the ability of players to reach the required pressures quickly and systematically, to produce the target values with limited feedback from aural and other bodily senses, indicates that they have intuitively memorized the relevant parameters of their instrument. This assumption was confirmed by Smith et al., (1990), who investigated the sensation of inspiratory volume and inspiratory pressure in wind instrument players. Their findings show that wind musicians have a considerably higher sensation for both inspiratory volume and pressure than nonmusical subjects. Fuks (1998b) applied a psychophysical production method to assess the sensation of blowing pressures in eight professional reed wind players. He found a proportional relationship between the measured blowing pressure and the estimated pressure level that expresses the players' perception. This approach showed that the subjects could judge quite accurately the multiplication of internal lung pressure, which seems to be a highly relevant sensory skill in wind performance. This particular ability may be trained by means of special exercises, for instance with the help of mechanical devices or a system controlled by a computer.

Lung Air Composition and Intonation

The temperature inside the instrument is related to intonation, so that an increase of 10°C (18°F) will raise the pitch by about a third of a semitone, or 31 cents (a semitone contains 100 cents).

Other factors may also affect intonation. Considerable variations in the concentration of carbon dioxide (CO₂) and oxygen (O₂) can be expected during playing. Fuks (1997) measured variations in the proportion of carbon dioxide in the air exhaled by players. This typically started at 2.5% and reached 8.5% at the end of long phrases. Meanwhile, the percentage of oxygen dropped from 21% to 12%.

The speed of sound depends on the gas properties and so proportionally affects the fundamental frequency of a wind instrument (excluding the free reeds). This can be demonstrated by having a wind player take a full breath of helium before playing: the tones may be transposed by an interval that exceeds nine semitones.

Under more realistic atmospheric conditions, the effect of the changes in gas composition, together with the change in air humidity, can produce a frequency decrease of up to 15 cents (15% of a semitone) during a long phrase. Trained players tend to compensate for this tendency for the pitch to fall by adjusting their embouchure and blowing pressure during their performance.

Vibrato

Vibrato consists of pulsations of pitch, usually accompanied by simultaneous pulsations of loudness and sound quality (Seashore, 1938/1967). Vibrato in winds can be produced by isolated or combined variations in blowing pressure or air-flow, restrictions in the airways caused by the movement of the larynx and tongue, and movement and tension of the lips and jaw (affecting the reed vibrations). In the case of the trombone, mainly in jazz but also in some classical players, vibrato may be obtained by the variation of the instrument's length. On the trumpet, a common type of vibrato consists of a gentle shaking of the instrument, which affects the embouchure. This technique is called hand vibrato.

In general, vibrato seems to include blowing pressure oscillation, be it in flutes (Fletcher, 1975; Gärtner, 1973), reed woodwinds (Brown, 1973; Weait & Shea, 1977; Fuks, 1998a; Secan, 1997), or brass. Some studies have observed oscillations of the vocal folds during vibrato, suggesting that they play a role as a control mechanism. For intense vibrato tones produced on oboe, bassoon, and saxophone, wide pressure oscillations have been observed, on average 10 cm H₂O for the oboe and bassoon and reaching values of 20 cm H₂O in some cases (Fuks, 1998a). These wide undulations cannot be produced merely by the laryngeal mechanism, which otherwise would produce vocal sounds simultaneously, but require a great deal of expiratory force.

Vibrato is seldom taught, in spite of its ability to enhance the player's sonority and expressivity. Many educators consider that vibrato should be discovered by the student. Some others prescribe a series of respiratory maneuvers, most of which involve pulsation in the respiratory flow, thought to facilitate the acquisition of vibrato. A possible reason for this pedagogical attitude is that vibrato involves such a complex neuromuscular coordination that a simplified or step-by-step method for its production might produce an unsatisfactory, artificial effect.

Respiratory Movements

Much attention has been given in the teaching of wind playing to respiratory techniques. Instructions on how to breathe in and out during sound production are usually given by the teacher, often while he or she is examining the student's body posture.

A few physiological methods are available to assess respiratory movement during wind performance. The technique called body plethysmography involves confining the player in a closed box, with only the head outside. Any variation in the body volume is detected and registered by the equipment. In spite of the

rather clumsy setup, this technique has been successfully applied to both singing and trombone experiments (Bouhuys, 1968). Another technique employs magnets and sensors connected to the player's thorax. These accurately record the respiratory movements. Magnetometers have been used during the playing of sustained tones in the flute (Cossette, 1993; Cossette et al., 2000). The latter study with three professional flutists found that they used similar movements when inhaling but quite different patterns of expiratory movements during expiration and that these differences did not result in perceptible differences in sound quality.

Respiratory Inductive Plethysmography (RIP) is a technique that also employs electromagnetic technology. Elastic bands that contain electrical coils are wrapped around the chest and abdomen of the player. These bands permit indirect measurement of cross-sectional areas (Strömberg, 1996). The variations of chest and abdominal areas reflect the variations in lung volume and significant changes in body position. In a study with eight professional players of the bassoon, clarinet, oboe, and saxophone, all performing different musical and respiratory tasks, the musical phrases were initiated at 55 to 87% and terminated at between 14 and 52% of the players' vital capacity, depending on instrument, piece, and phrase length (Fuks & Sundberg, 1999b). Vital capacity is the maximal air volume that a person can exhale from full lungs. The players generally showed simultaneous and in many cases equally important contributions from rib cage and abdominal wall during playing. These findings often contradicted the players' own idea of how they used their respiratory apparatus. Some of the players even claimed that their rib cage was not moving at all during inhalations and playing, while the technical equipment demonstrated quite conclusively that their rib cage was responsible for more air displacement than the abdominal wall.

The studies mentioned earlier, undertaken with professional players of the flute and reed woodwinds, show that there is no single correct way to make expiratory movements in wind playing. We nevertheless recommend a relatively constant, systematic, and automatic expiratory pattern, as this allows the player to focus on other aspects of performance, such as blowing pressure and embouchure control.

Pedagogy and Winds

Wind playing imposes high demands in terms of learning and handling respiratory techniques, arm/hand/finger movement, tongue articulation, body posture, and embouchure control. Physiological aspects are therefore extremely important. However, it is also important for players to understand something of the acoustical behavior of the instrument they play, in terms of the function of the mechanisms and practical factors that produce the sound. In addition, psychological issues are involved, such as the aesthetic dimension of music and the emotional stress induced by intensive practice and public performance. Therefore, it is not uncommon that wind players inherit and develop an informal investigative attitude toward the physical, physiological, and psychological phenomena that they experience and try to apply these principles in teaching and

reasoning on the art. In some cases, these ideas are grounded in scientific knowledge but also include a combination of empirical conception and mental images. These images are supposed to represent an ensemble of complex tasks and sensations, which can hardly be expressed objectively (Kohut, 1985).

Physical, physiological, and psychophysical variables are continuously coordinated during performance and must be trained through systematic and repeated exercises, usually with the support of method books and under the supervision of an experienced teacher. The usual approach of most teaching methods is to combine the performance of musically significant studies with scales, arpeggios, and other drills that are used to hone the playing of complete works. Modern teaching tends to emphasize musical expression, even in studies that are relatively mechanical.

Many pedagogical methods briefly consider techniques for respiration and breathing for sound production and sometimes the underlying acoustical principles. In some cases, these methods review some basic anatomy of the respiratory apparatus, sometimes complemented by descriptions of the inspiratory and expiratory muscles, recommendations on posture during playing and on optimal inhalation strategy, and instructions on how to achieve “support” and “use the diaphragm.” Some educators suggest that these terms may evoke an image of stressful playing that has led to a number of breathing and embouchure problems, even with experienced performers (Fadle & Kohut, 1999; Kohut, 1985).

Arnold Jacobs (1915–1998), a brilliant tuba player and teacher, spent his entire professional life studying respiratory and psychological aspects of winds. Jacobs was probably the most influential wind instrument teacher in the United States and other countries, particularly among brass players (Stewart, 1987; Frederiksen, 1996; Kohut, 1985). In his lessons and workshops, Jacobs used an ensemble of devices, developed for respiratory clinical purposes, to provide visual feedback to his pupils that regarded pulmonary pressure, airflow, and lung volume. Interestingly, Jacobs used a terminology that was more pedagogically oriented than based on physical facts. For instance, he emphasized *flow* as a key word for playing control, as opposed to *pressure*, which he deemed as a somewhat negative term. Because of their psychological effect, different terms tend to trigger different behaviors in players. There is another reason that Jacobs used the term *flow*. In brass instruments, a wide range of combinations of mouth pressure and lip resistance produce the same output level (Weast, 1965). If the player is able to use configurations that require lower pressures, such as lower embouchure resistance and/or stiffness, then the airflow should be maximized and the playing effort reduced (Nederveen, 1969/1998). These general principles are likely to apply also to other winds.

Final Remarks

Knowledge acquired through scientific research cannot replace, and should not challenge, the rich imagery that inspires and guides the artist. But it may help to question, consolidate, and fill the gaps left by individual observation, thus providing an objective background for education in wind playing. Without doubt,

some research results conflict with common practice and concepts among the profession, which are often based on obsolete hypotheses that have gained credibility among musicians over many generations.

However, because measurement methods and experimental procedures are still coarser than the musical subtleties involved in performance, it is important for researchers to understand the artists' reality and to take advantage of their skills and long-term experience when designing musically relevant investigations. Collaboration among areas such as physiology, acoustics, psychology, engineering, and musicology will help to enhance knowledge of music performance in winds. As a first step, there is a need to develop some consensus concerning the meaning of technical terms, in ways that promote a better understanding and effectiveness of their application to music teaching and learning.

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